

MIOS – Market Intelligence Operating System (Easy Skill Australia)

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Declaration

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Date: 03/05/2026

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Executive Summary

This project has been proposed for Easy Skill Australia, a workforce solutions company that operates in Australia, Papua New Guinea and few other areas. It specializes in recruiting and providing technical specialists in projects related to various areas such as mining, oil & gas, construction, defense, and energy. The company's core business revolves around understanding market demand. This means knowing which companies are hiring right now, what skills are sought for by them, and when new projects are emerging. Identifying new projects is crucial to ensure the right people are sent to the right places at the right time.

Although Easy Skill operates in a highly specialized market, it currently lacks a proper system that can capture and analyze market intelligence. That's why it lacks behind in many aspects such as gathering information about current industry trends, hiring activity, etc. Most insights it gets are from human-based processes, like searching through job boards (websites where jobs are posted by different needful companies), LinkedIn and some internal systems used by the company such as HubSpot and Teamtailor. This disoriented and fragmented approach has led to many issues in the workplace.

To address these challenges faced by Easy Skill, this project proposes the development of Market Intelligence Operating System (MIOS), a system that will be designed to collect information automatically from various online sources, observe and analyze it, and deliver valuable insights to the company so that they can take robust actions based on these insights. It's a python based multi-agent AI system that integrates internal and external data sources and converts them into actionable intelligence.

To ensure insights reach the business development team, the system deploys the outputs generated through Slack and Google Docs. Likewise, the MIOS operates on three modes namely Mode Monitor, Mode Push and Mode Publish. Proof of concept (PoC), a small sample like structure is also developed which includes collecting data, storing such data in database and analyzing those data with the help of AI tools, generating needful insights to the company and delivering them via Slack Digest like a newsletter. Key Performance Indicators (KPIs) such as accuracy, cost efficiency, availability, etc. are used to monitor how effectively the system is running. These KPIs are crucial for assessing quality factors or non-functional requirements of any developed system.

MIOS is designed to be implemented in phases, starting from key markets in Australia and Papua New Guinea and slowly transitioning to other areas as we move on. This will help with controlled automation and gradual enhancement in operational efficiency and decision making.

1. Introduction

1.1. Business Context and Problem Statement

Easy Skill Australia is a recruitment and workforce solutions company which works to provide highly skilled technical professionals to various industries and engineering projects. It operates in multiple regions around the globe including countries like Australia, Papua New Guinea, New Caledonia, France, Madagascar, and Singapore. The company specializes in providing skilled workers, hiring them for permanent roles and managing workforce solutions based on some emerging projects.

Because of the area of operation of Easy Skill, it's evident that it relies heavily on market intelligence. It's important for the company to get early insights regarding industry activity and opportunities. Identifying early indicators of project announcements, hiring increases, and new industry developments is crucial to stay ahead in the game. However, Easy Skill lacks a real-time structured system that can capture and analyze this intelligence. The source of observation is dependent on manual review of job boards (websites that have vacancy posts), reading industry news regularly, observing LinkedIn activities and profiles manually. These manual and outdated mechanisms don't provide quick analysis of market trends and hence can lead to missed opportunities.

Apart from the external data sources, Easy Skill has its own set of system that it uses such as HubSpot CRM and TeamTailor (a software used to manage interaction with clients). This helps in collecting valuable internal information. Various useful insights into client calls and interactions, conversations with candidates, and activities on recruitment are stored by these systems. However, data stored here is not systematically analyzed and optimally analyzed, which means that a major portion of business intelligence still remains unused.

Another challenge observed by Easy Skill is its inefficiency in matching existing consultants to prevalent market opportunities. When consultants finish an ongoing project, the BD team has to manually find them a new project by scanning different job boards, contacting clients one by one, and constantly reviewing market information for any availability. This process results in extra time consumption and inefficiencies, although opportunities may already exist in the available data itself.

The challenges observed have an impact on both operational efficiencies and strategic advantages for the company. Valuable time and resources are spent on tedious manual tasks instead of high-value activities. Also, market signals that come from smaller segments of the market, where timing plays a big part, is missed. Upon observing these challenges, Easy Skill has opted to build an automated system that can continuously collect, analyze, process, and deliver market intelligence to all the stakeholders in a usable format. These requirements form the foundation for our MIOS project.

1.2. Goals, Objectives and Success Criteria

The primary objective of our project is to design and develop the Market Intelligence Operating System (MISO), a system that can automate the process of collecting, processing and delivering insights of market intelligence for Easy Skill. The system uses modern technologies such as Google Cloud Platform for cloud computing services, BigQuery for cloud database management and Anthropic Claude API for smart text analysis and pattern identification. These technologies help in handling large volumes of data that help to generate meaningful insights for the company.

New automation and digital technologies have been helping to transform work and economy. AI can be termed as one of the tools in the latest toolkits of automation that perform tasks with less cost, faster and better (Tyson and Zysman, 2022).

The objectives of the project are divided into separate distinct phases. The first objective is analysis of the industry and technology landscape. This involves diligently evaluating existing tools such as AI frameworks, available data scraping tools and possible data storage solutions.

The next objective of the project is designing the architecture of the system (a blueprint that shows how different components of the system interact with each other). This objective revolves around defining how data flows from external sources into the company's database storage systems and then into different AI components for processing. Integrating AI into administrative procedures can help to revolutionize how processes are conducted and can basically change the established forms in which actions are taken. However, implementing them requires a comprehensive evaluation of strengths and weaknesses of different systems (Parycek et al., 2024).

The third and very important objective is to develop proof of concept (PoC). This will act as a working prototype that shows the core functionality of the system upon which further buildup can be achieved. This includes the following:

- Data collection from any job board
- Data storage into BigQuery
- Classification of signals such as identifying patterns of hiring trends using AI
- Delivery of insights through Slack

The final objective of the project is to evaluate the system with the help of Key Performance Indicators (KPIs), measurable criteria that are used to assess performance of any system. Some criteria include accuracy, processing time, cost efficiency, and the quality of the output.

1.3. Project Scope

The scope of this project can be categorized into two parts. It includes both the design of the full working system and the development of a small proof of concept (PoC). Full MISO system consists of three modes:

- Mode 1 – Monitor: An automated system working on autopilot, that scrapes external sources, reads internal conversation notes, and produces intelligence every week on a Monday morning Slack digest.

- Mode 2 – Push: This mode works with already existing consultants or candidates. The BD team submits the person's profile into MIOS, MIOS searches the market and finds out companies that are in need of someone with consultant's skills. Simply put, it finds matching companies or opportunities by taking a consultant's profile.
- Mode 3 – Publish: Mode Publish turns the market intelligence into reports that are useful for clients and into thought-leadership content. For example: a quarterly report containing details about mining hiring trends in Australia and Papua New Guinea. So, mode publish doesn't just understand the intelligence but rather changes them into actionable reports for the clients.

The PoC is however narrower in scope. It just demonstrates the system's core functionality by collecting data from a single job board, structured storage into BigQuery, classification of AI logic and delivery of automated reports. The project also includes analysis of various technology options, initial set-up and ongoing cost, feasibility assessment and some recommendations for future implementation and improvements.

The PoC doesn't include certain features such as full integration with HubSpot and Teamtailor, use of real data, deployment into the company's live system and advanced features such as live processing. To reduce cost and complexity, the system runs once per week instead of continuously.

The key assumptions held are availability of synthetic data (data that are just used for testing purposes), access to Easy Skill's cloud services, and alignment with the existing legacy systems and technology. The challenges or constraints here are limited time of the project, limited contact with the mentor for intricate details, budgetary restrictions, and also the need to deliver a PoC within a semester.

2. Project Planning

This repository contains all project management artefacts associated with the MIOS — Market Intelligence Operating System project undertaken by a team of 5 students and delivered to Easy Skill Australia for ICT942. Artefacts within this repo including Project Charter, Gantt Roadmap, RACI Matrix, Risk Register, Feasibility Study and Sprint Plan outline how this 12 week project will be governed and delivered.

2.1. Project Charter

Project Title: MIOS — Market Intelligence Operating System: Architecture & Mode Monitor Proof of Concept

MIOS project is a 12-week semester-long industry experience project sponsored by Pierre Bussy, Co-Founder of Easy Skill Australia (Strategic Accounts & Growth) delivered through Ready grad. The team of 5 students will create, prototype, and assess an AI-driven market intelligence tool that powers automated extraction, categorisation, and delivery of hiring/project signals across mining, oil & gas, construction, defence and energy industries for Easy Skill's business development team.

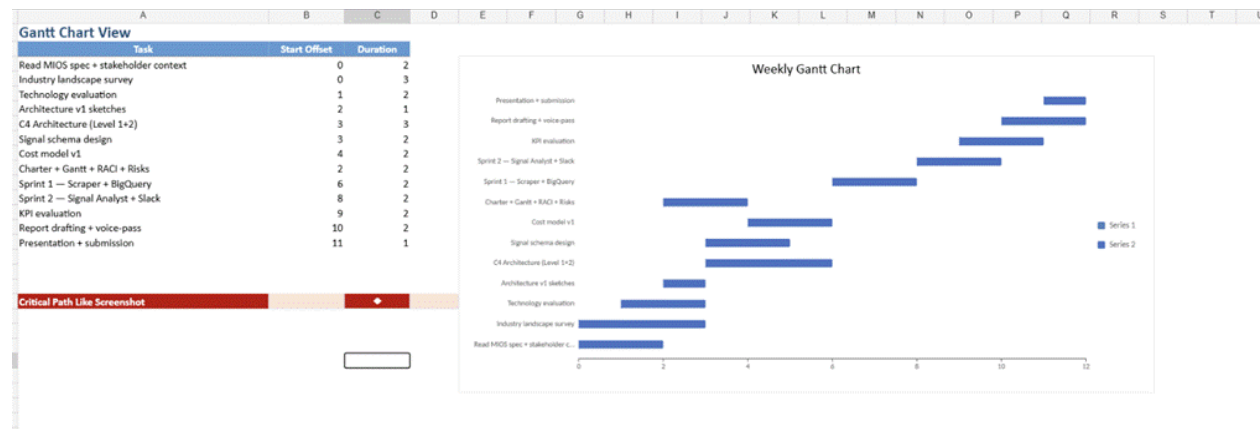
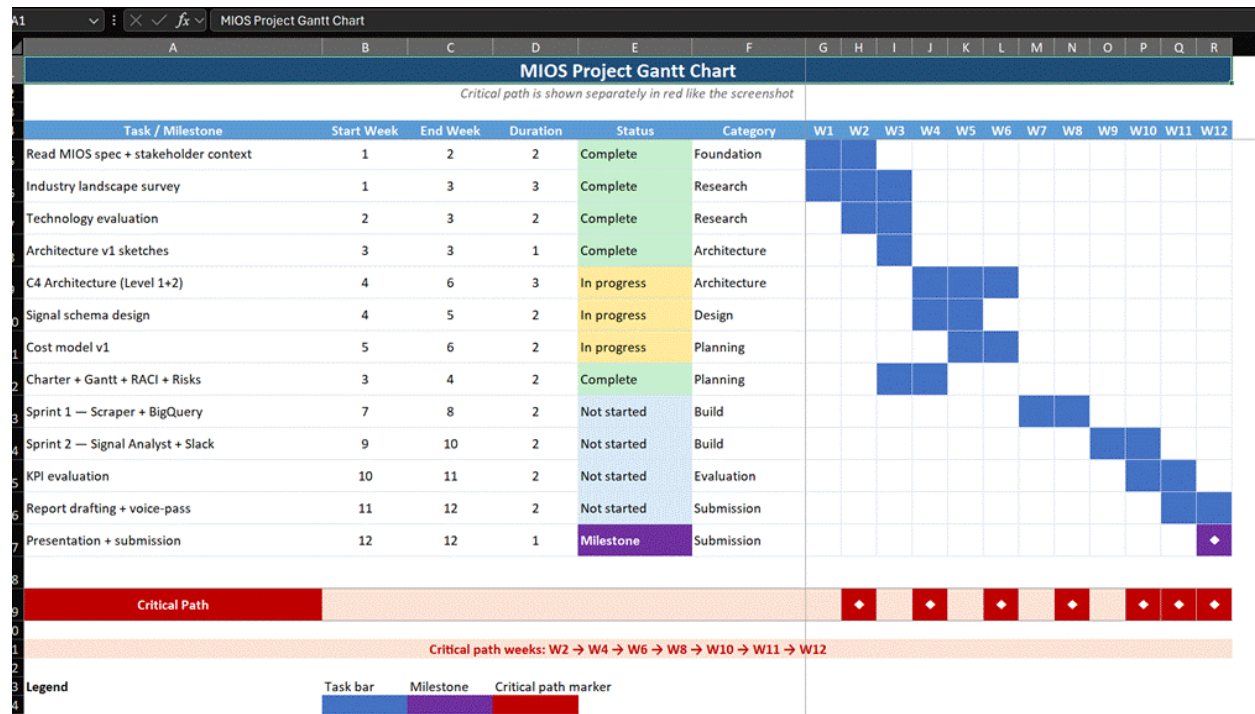
Role in project: The student assigned Prashant as project manager and will Project Lead / Editor and PM Artefacts Lead. He will lead the weekly team sync and have ownership of voice-pass and all PM artifacts. Timeline: Weeks 1–12 of Semester 1, 2026 (up until submission date). Budget: \$300 agreed AUD budget to spend on cloud/ API usage (GCP free tier, Anthropic Claude API, Apify free tier). Labour is sunk cost given this is a unit. Stakeholders: Too many to list! Key stakeholders are Pierre Bussy and Christopher (Co-Founders, sponsors + business validation), Jonathan (BD Lead, and main system user), Easy Skill BD Team (day-to-day users of the system), and ICT942 Unit Coordinator (Teacher).

The team has decision-making authority on: Cloud/ API spend of up to \$300; Choosing tooling from within Easy Skill's tech stack (Anthropic, GCP, Apify, Slack); Determining what synthetic dataset is used to test/evaluate the PoC. The team does not have authority to use real Easy Skill Hubspot or Team tailor data.

Deliverables must meet the following standards: Criteria for acceptance: meet all 4 SMART objectives (O1–O4) or clearly documented why not; GitHub repository of PoC is available with a working README file.

2.2. Project Roadmap — Gantt Chart

The project consists of four phases completed in 12 weeks. Cells that are shaded show when work is occurring for each task. The critical path is highlighted with ♦ symbols in the last row.



Phase 1 (W1–W3) is comprised of discovery related work: reading the MIOS spec, conducting industry landscape survey, and creating initial architecture sketches. Phase 2 (W4–W6) culminates in the completed design: C4 architecture at both Level 1 and Level 2, signal schema, cost model, and all supporting PM artefacts. Phase 3 (W7–W10) executes the PoC. This work occurs during 2 agile sprints — Sprint 1 covers scraper + Big Query, and Sprint 2 covers Signal Analyst agent + Slack delivery. Finally, Phase 4 (W10–W12) encompasses evaluation work, drafting of the report, rehearsal of the presentation, and finally, submission.

The critical path (shown in red): W3 landscape survey done → W6 architecture reviewed by peers → W8 scraper/Big Query working → W10 full end-to-end Slack digest → W11 KPI's evaluated → W12 report turned in.

Contingency: If Sprint 1 (W7–W8) falls behind schedule, the PoC scope is reduced to 'scraper → Big Query → CSV output'. The Stretch goals of building the Signal Analyst agent and delivering the data to Slack are removed. This decision is made by M5 no later than the end of W8. Reasons for the deferral are captured as a scope change in section 5.2 lessons learned.

The RACI matrix assigns clear ownership for every major deliverable. Rules: every deliverable has exactly one A (Accountable) and at least one R (Responsible). C = Consulted (input expected before completion). I = Informed (kept in the loop but no input required). Where A and R coincide, the notation A/R is used.

Deliverable	Prashant Project Lead	Addit PM Artefacts	Bijaya Industry Analyst	Banish Solutions Arch.	Shishir PoC Engineer
Charter, Gantt, RACI, Risk Register	I	A/R	C	C	I
Industry & Tech Landscape Analysis	C	I	A/R	R	C
C4 Architecture Diagram + DFD	C	I	C	A/R	R
Signal Schema (BigQuery)	I	I	C	A/R	R
Technology Stack Justification	I	C	C	A/R	R
CAPEX/OPEX Cost Model	C	R	I	A/R	I
PoC Sprint 1 (Scraper + BigQuery)	I	I	C	R	A/R

PoC Sprint 2 (Signal Analyst + Slack)	I	I	C	R	A/R
Synthetic Dataset (~200 postings)	I	I	A/R	C	R
KPI Harness & Evaluation	C	I	R	C	A/R
Executive Summary, Intro, Conclusion	A/R	C	C	C	C
Final Report (assembled)	A/R	R	R	R	R
Presentation Slides	R	R	R	R	A/R
Live Demo + Backup Recording	I	I	C	C	A/R

Pairing dependencies: Bijaya and banish will sync up weekly. Architecture Banish relies on the gap analysis from Bijaya. banish and Shishir will sync weekly. Architecture dictates what goes into Shishir's build of PoC. addit and M4 will reconcile the Cost model section and Feasibility sections. Bijaya and Shishir work together on a synthetic dataset. Prashant will read each section before being checked off as complete. Pairs create no siloed section writing. The final product will mesh coherently across all sections.

2.3. Risk Register

The risk register uses the standard 5×5 likelihood–impact scoring matrix. Rating thresholds: Low < 6, Medium 6–11, High 12–19, Critical 20–25. Response strategies follow the standard four-option model: Avoid, Transfer, Mitigate, Accept. All High-rated risks (score ≥ 12) have mitigation actions that are tracked in the weekly team sync by M1.

ID	Risk	L	I	Score	Rating	Response	Owner
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R01	Job board blocks scraper / switches up HTML during crawl	4	3	12	High	Avoid — leverage low-friction source (PNGworkforce / public RSS feed); use synthetic-only dataset upon blockade	Shishir
R02	Anthropic hit API usage limit mid-assessment	3	4	12	High	Mitigation: Set spending limit to A\$200; default to Haiku only; cache LLM calls locally during dev	Shishir
R03	Synthetic dataset doesn't contain representative signals → KPIs inaccurate	3	4	12	High	Avoid/mitigate — M3 is constructing dataset from actual public responses (anonymised); mixture of 3 sources; M4 validation pass for realism	Bijaya
R04	Team member out for Sprint 1–2	3	4	12	High	Mitigate — RACI structure in place to provide coverage; created shared GitHub repo in W1; Prashant reshuffled work within 1 day	Prashant

R05	Signal Analyst classification confidence <80% (KPI miss)	3	3	9	Medium	Accept partially — clearly record situation in retrospective and lessons learned; refining prompts in Sprint 2	Shishir
R06	Slack workspace / webhook permissions denied	2	4	8	Medium	Mitigate — created personal Slack workspace in W1; webhooks aren't paid features; fall back to printing to console	Shishir
R07	Cloud costs exceed A\$300 spend limit	2	3	6	Medium	Plan by setting billing alerts at 80% threshold; Use GCP free tier services, Apify free tier; Use serverless computing resources only	Addit
R08	Presentation/demo does not work at the time of presentation	3	3	9	Medium	Plan ahead by having recorded walkthrough of full backup; practice using backup slides that can be shown offline	Shishir

R09	Ambiguities in Easy Skill spec (skill taxonomy undefined, etc)	3	2	6	Medium	Plan ahead by documenting assumptions in 1.3; make explicit callout in W1 framing memo to PC coordinator	Bijaya
R10	Design does not meet requirements of 'C4 Level 2' checklist	2	4	8	Medium	Plan ahead by having members of M4 review C4 model documentation by W4; complete peer-review by W6	Banish

Primary risks: four risks rated at 12 (High) - scraper getting blocked (R01), API quota running out (R02), synthetic dataset being unrealistic (R03) and team capacity (R04). All four have associated mitigations which are actively being pursued. The two highest individual impact risks (R02 and R03 with I=4) are owned by M5 and M3 respectively. The risks will be reviewed every week in our weekly sync. No risks are currently ranked as Critical (20–25) as the scope of this PoC (single source, using synthetic data, not deploying into production) caps the severity.

2.4. Feasibility Study

The feasibility study looks at the MIOS project through four standard lenses. Each lens comes to a definitive Go/No-Go decision. All four decisions are GO - the project is feasible given the scope and constraints.

Dimension	Assessment	Verdict	Key Caveat / Condition
Technical	Python (including all integrations), Anthropic Claude API (which has great documentation), GCP BigQuery + Cloud Run (on free tier), Apify (used successfully in Easy Skill's PNG prototype), Slack webhooks. Demoable prototype already in production at Easy Skill - has been proven end-to-end feasible.	GO ✓	Primary risk: Signal Analyst prompt engineering - will be worked out through iteration during Sprint 2 (R05). Team has skills in all needed areas.

Operational	The proposed PoC works with a five-person team for twelve weeks. This should be sufficient, given the current scope of the PoC. Easy Skill have already shown willingness to engage through the previous engagement. Architecture wise, we are not writing back to HubSpot/Teamtaylor automatically, avoiding that complexity.	GO ✓	Production will need capacity at least equal to S5 devs to operate maintainer capacity-highlighted under 6 reqs.
Economic	PoC cost: A\$300. Production opex: A\$230–A\$630/month. Avoid one missed placement (~A\$30K–A\$80K fee), and you've covered 12 months of opex. Bullhorn GRID Reference: 18–25% lift in placements from curated market intelligence.	GO ✓	Full year cost estimate ~A\$121K. Savings from manual scanning effort alone (~A\$135K) pays for Year 1 in 6 months.
Legal & Regulatory	Privacy Act 1988 (Cth) / Australian Privacy Principles (APPs) were deemed satisfied by the scoped Proof of Concept (PoC) as it only uses synthetic-data. The scraped job board data is publicly-available information. There is no Personally Identifiable Information (PII) of real candidates or clients used in this project.	GO ✓	Deployment into Production will necessitate a Privacy Impact Assessment in accordance with APP 1. Google Cloud Platform (GCP) region was locked to ap-southeast-2 (Sydney) to comply with data sovereignty requirements.

Overall feasibility conclusion: All four dimensions are GO. The binding constraint is signal classification accuracy in the PoC — mitigated via R03 (realistic synthetic dataset authored by M3) and R05 (prompt iteration during Sprint 2). The project's economic case is particularly compelling: labour savings from manual scanning alone (~A\$135K/year) would cover Year 1 total cost (~A\$121K) within the first six months of production operation, before any placement uplift benefit is counted.

The feasibility study does not address Mode Push (Profile Matcher) or Mode Publish (quarterly report generation), as these are explicitly out of scope for the PoC per §1.3. Their feasibility is implicitly validated by the PoC architecture — they share the same pipeline pattern.

3. Industry Analyst

3.1 Current State Assessment

Currently, Easy Skill Australia works in a fragmented, mainly manual market intelligence environment. LinkedIn activity and job boards (SEEK, PNG Workforce, etc.) and internal CRM software (HubSpot, Team Tailor) are where the BD team sources most of their hiring signals and new project opportunities.

This process is manual and messy. It leads to delayed information (Buttle & Maklan, 2019), incorrect data gathering and lost opportunities. By the time we see hiring spikes in some markets such as mining/oil and gas (Laudon & Laudon, 2020), it's too late and we lose our advantage.

The present workflow of the system can be stated in terms of three key actors:

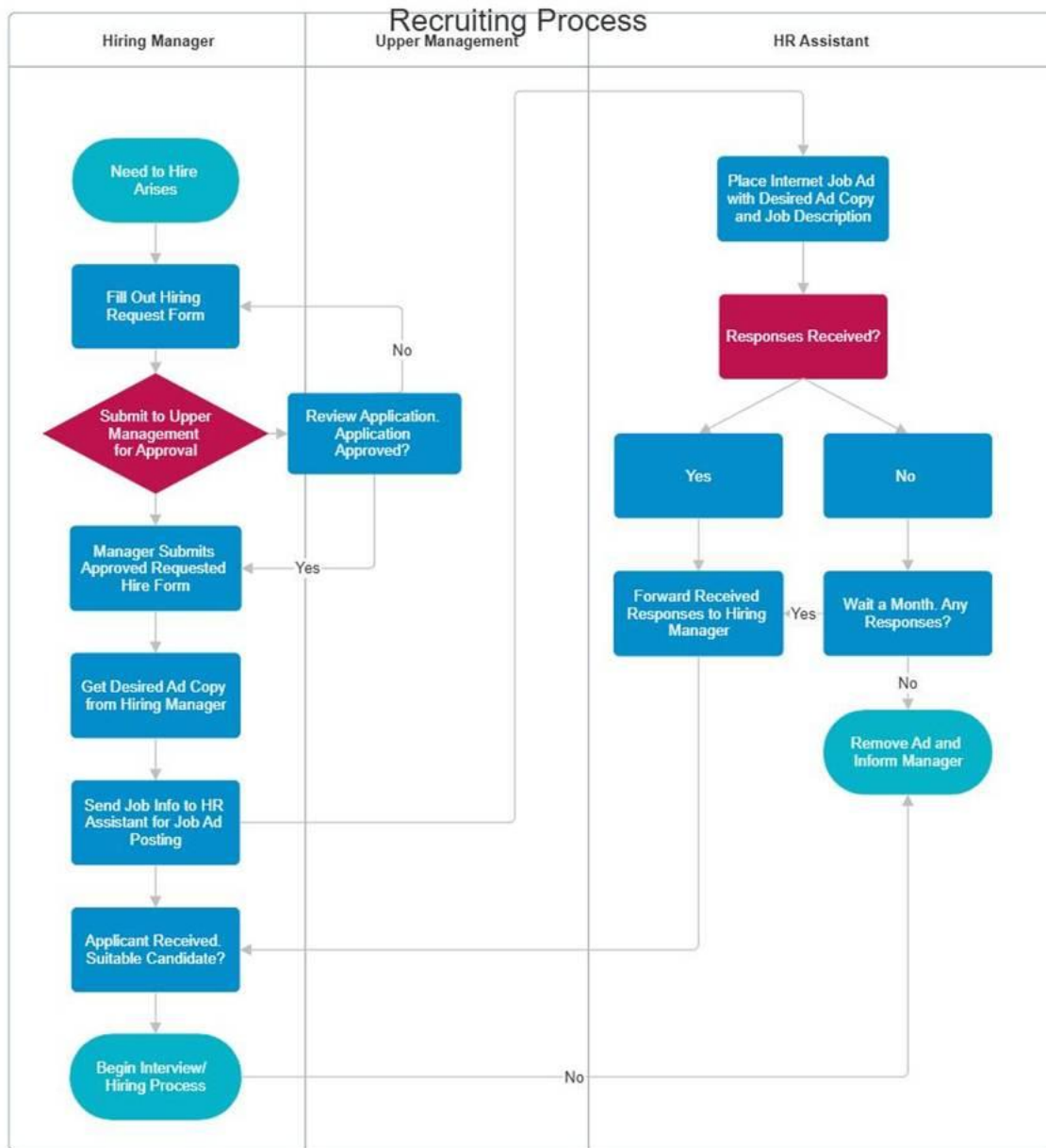
- Source: Job boards, LinkedIn, industry news
- BD Team: manual scans, filters & data interpretation
- Internal Systems: HubSpot & Team customize store interaction data, but no analytics integration

Key	Issues:	Current	Status
	• Manual Effort: time-intensive scanning of multiple platforms • Data silos: external and internal data not integrated • No real-time insights: weekly or ad-hoc awareness versus continuous monitoring • Manual matching: Consultants are manually matched to opportunities • Unexploited CRM data: Valuable insights that are not being used		

A little taste of automation is a PNG-specific Zapier prototype that damages job board data and delivers a weekly report via Slack. However, this solution is not scalable to Easy Skill's larger scope of activities, is limited to one area and one source and has no analytical capabilities.

Swimlane Diagram

Data Source Layer	BD Team Actions	Internal Systems
Job boards scraped manually	Search for jobs daily	HubSpot stores client interactions
LinkedIn browsing	Identify hiring trends	TeamTailor stores candidate data
Industry news reading	Interpret signals manually	No analytics performed
Existing contacts / personal network	Contact clients manually	No automated matching



This fragmentation suggests the need for a centralised, automated Market Intelligence Operating System (MIOS) that continuously combines and processes data.

3.2. Technology & Market Survey

To build MIOS, we must take inventory of the current technologies available for data collection, storage, AI processing, and distribution. These solutions represent some of the industry standard practices that companies use today for recruiting intelligence and data-driven decision-making.

1. Apify

Apify is a web scraping and automation platform commonly used to scrape structured data from job boards and websites. Offers premade actors and scales to crawl. (Google Cloud, 2024).

- Strength: Scalable, easy to scrape
- Limitation: Must be maintained if the website changes structurally.

2. Playwright (Python/Node.js)

Playwright is an open-source library that provides browser automation. It's great for scraping dynamic websites.

- Strength: Can handle javascript websites
- Limitation: Takes more initial work to configure over SaaS solutions

3. Google Big Query

A fully managed, scalable data warehouse for analytics. • Strengths Very scalable, quick SQL queries • Limitation: Cost rises with heavy usage

4. Claude via Anthropic's API

Large language model for text categorization, summarization and signal detection • Strength: Sophisticated NLP to classify market signals

- Limitation: API cost and timely quality

5. Webhooks for Slack

Delivers real-time analytics directly to teams.

- Pro: Realtime connections/integrations
- Con: Minimal formatting/analytics options

6. Cloud Run (GCP)

Fully managed serverless platform that runs containerised applications.

- Pro: Inexpensive and scales easily
- Con: Experiences cold start delays

7. Cloud Scheduler

Schedule pipelines to be triggered.

- Strength: Workflow automation
- Limitation: Restricted to time-based triggers

8. HubSpot / Team tailor internal CRM and recruitment tools.

- Strength: Abundant internal data
- Limitation: Not intended for analytics or intelligence creation

Summary Insight

The poll demonstrates that no instrument provides end-to-end market intelligence.

Modern systems use instead the following:

- Scraping tools → Data gathering
- Cloud storage – Data centralization
AI models -> Generation of insight
- Messaging tools → Distribution

This scenario justifies the requirement for MIOS as an integrated multi-agent system.

3.3 Comparative Analysis Table

The comparison indicates that while there are several tools throughout AI, scraping and data infrastructure layers, the best approach for MIOS is not a single platform but a well-integrated combination of technologies. The Anthropic Claude API (Anthropic, 2024) is a powerful AI framework with natural language processing capabilities and the ability to generate structured results, which makes it a good choice for signal classification tasks.

Conversely, LangChain and CrewAI offer these multi-agent orchestration features, but at the cost of additional complexity and framework dependency that isn't required for a batch-style pipeline like MIOS will be.

Additionally, Apify offers a practical advantage for the data collection layer as we have already validated in the Easy Skill. Cost may come into play at scale, but at least for now Apify is a valid

choice. Google BigQuery is the clear winner for data storage/analytics due to its scalability and native integrations with Google Cloud products.

In summary, this analysis shows that the technology stack proposed in the MIOS architecture (Anthropic + Apify + BigQuery + Slack) is feasible from a technical standpoint as well as efficient for operations. The main tradeoff is making the choice for simplicity/maintainability over migrating to more complex frameworks with more features.

Tool	Category	Strengths	Weaknesses	Best Use Case
Apify	Scraping	Easy to use	Paid tiers, site breakage	scraping job boards
Playwright	Scraping	Handles more dynamic sites	Technical setup	Custom scraping
BigQuery	Data Storage	Scalable, fast queries	Cost with heavy queries	Analytics warehouse
Claude API	AI/NLP	Strong text understanding	Cost, prompt tuning	Signal classification
Slack Webhooks	Delivery	Real-time alerts	Limited analytics	Insight distribution
HubSpot	CRM	Stores internal data	No analytics intelligence	Client interaction tracking

3.4 Gap Analysis

Gap 1: Absence of Automated Data Collection

Current: Manually scan job boards

To-Be: automatic scraping (Apify)

Impact: Saved A\$135K/year

Verification: PoC scraper -> BigQuery

Gap 2 – Lack of Signal Classification

As-Is: Raw data manual interpretation

To-Be: AI classification (Claude)

Impact: Faster, more consistent insights
Verification: KPI ≥80% classification accuracy.

Gap 3: No Data Integration
As-Is: External + internal data siloed
To-Be: BigQuery system centralised
Impact: Facilitates cross-analysis
Verification: Architecture of data pipeline
Gap 4: Not Leveraging CRM Intelligence

As-Is: No use of HubSpot/Teamtailor for insights
To-Be: AI extraction of conversation data
Impact: Gaining a competitive edge
Architecture verification: Validation

Gap 5: Manual Consultant Matching
Manual placement search As-Is
to Be: AI-based matching (Mode Push)
Impact: Enhanced placement efficiency
Verification: Next phase (not PoC)

Gap Summary Table

Dimension	As-Is	To-Be
Data Collection	Manual	Automated
Storage	Fragmented	Centralized
Analysis	Human	AI-driven
Insights	Delayed	Real-time
Decision-making	Reactive	Proactive

4. Solution Design

4.1 Proposed Architecture

MIOS has been structured on a single guiding theory: extract signal from many sources, classify it in one structured workflow, and surface the result where the Easy Skill Australia team had already worked i.e. Slack. There is no manual handoffs between data receiving and intelligence appearing and no new tools to adopt.

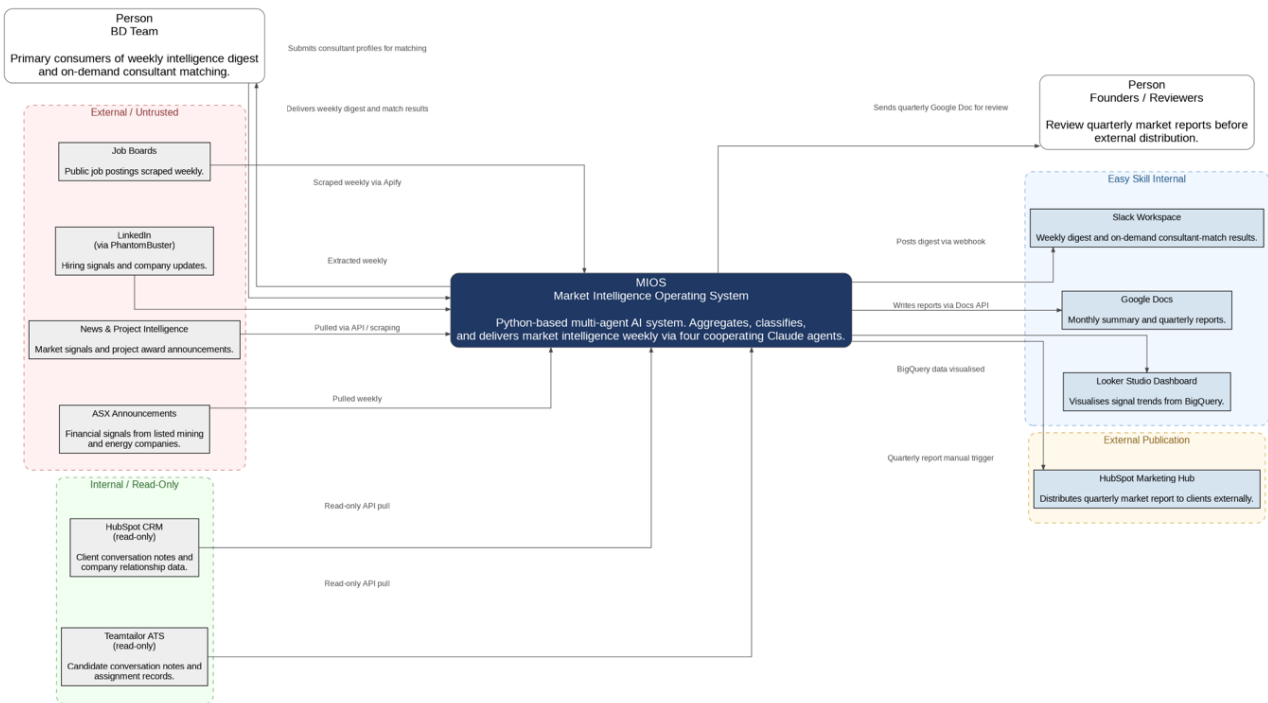


Figure 4.1: MIOS System Context — C4 Level 1

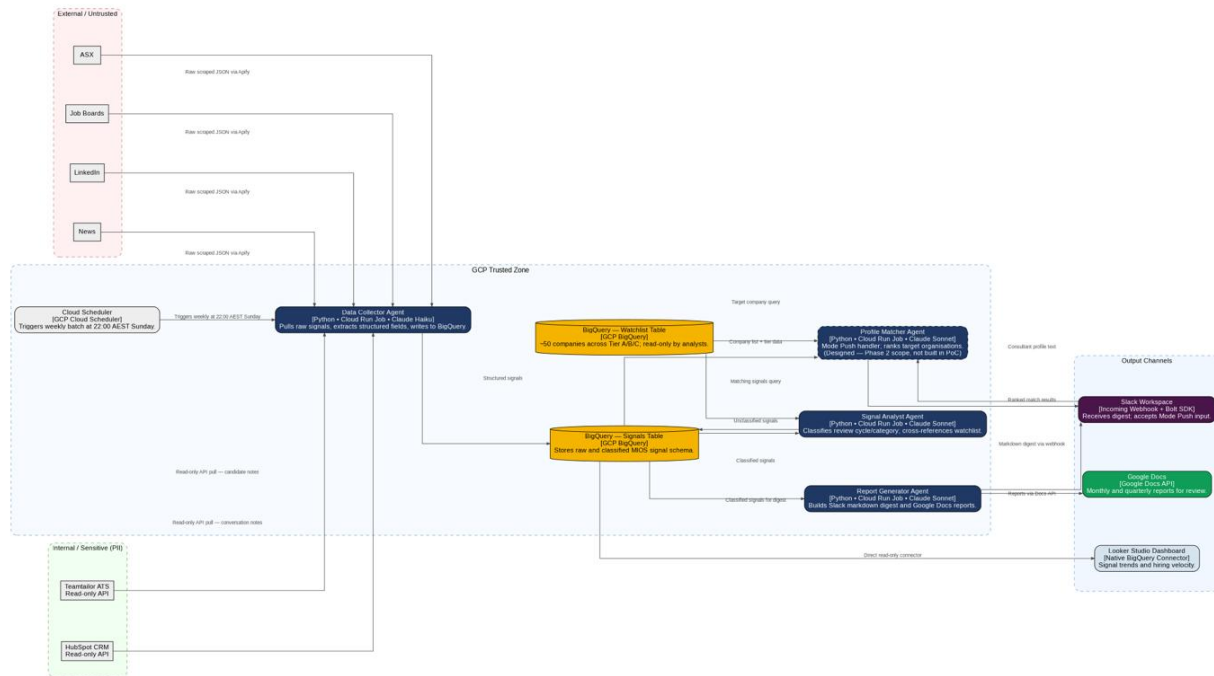


Figure 4.2: MIOS Container View

Weekly Batch Cycle (Sunday night to Monday Morning)

The data collector agent will activate by cloud scheduler at 22:00 AEST on Sunday. Agents will pull from configured Apify actor and API connector i.e. on job board in the POC, more than eight sources categorized in production (job boards, LinkedIn via PhantomBuster, news feeds, ASX announcements, project intelligence database, and read only pulls from HubSpot and Teamtailor). Claude Haiku manages field extraction in this layer which is the cheapest model in the Anthropic tier and the task is mechanical: extract company name, role title, posting date, location and role category from raw text, then generate structured row into BigQuery. Before calling any LLM, eliminating duplicates and clearly out of sector entries a Python pre-filter will run. This single step reduced estimated API spend by 60% in comparison to an LLM-first approach (Anthropic, 2026).

Signal Analyst (Classification and Pattern Detection)

When signals appear in BigQuery, the Signal Analyst agent i.e. Claude Sonnet retrieve the new rows, cluster each on two axes i.e. review_cycle (weekly, monthly, quarterly) and signal category (hiring velocity, project, competitive market intelligence, financial, leadership) as well cross references the watchlist table (50 companies in three tiers). Signals which match no watchlist entry are flagged as emerging opportunities for Business Development review. Categorized results are written back to the signal table. Sonnet is used here, not Haiku because multi-axis categorization and pattern reasoning require greater model capability (Anthropic, 2026).

Report Generator, Mode Push, and Mode Publish

The **Report Generator** (Sonnet) ingests the weekly signals from BigQuery into a Slack-formatted markdown digest of Signals: Key Signals, Market Pulse, Hiring Velocity, New Names. A formatted report lands in Slack by 08:00 Monday morning through an incoming webhook.

Mode Publish is similar except runs on quarterly cadence and ends at a Google Doc which must be reviewed by a founder before external distribution through HubSpot Marketing Hub. That manual review step is architecturally enforced: there is no automated send action that publishes externally available content (spec §8.3).

Mode Push exists in architecture but was only built Phase 2 (see §1.3 -- explicitly excluded from PoC).

Key Design Decision

- Read from many, write to few i.e. agents never write back to HubSpot or Teamtailor, only to BigQuery and delivery channels.
- Weekly batch, not daily i.e. market signals applicable to Easy Skill do not move faster than a week; daily runs would replicate API cost sevenfold for no intelligence gain
- Haiku for extraction, Sonnet for analysis i.e. split model architecture keeps the weekly batch inside the \$5 PoC cost target while maintaining classification quality.
- Custom Python orchestration over a framework — covered in §4.2 justification narrative.
- Human-in-the-loop on all external outputs i.e. the quarterly market report entails founder sign-off before distribution; the pipeline enforces this by design, not by policy.

4.2 Technology Stack and Justification

The basic stack—Google Cloud Platform, Apify, Slack, and the Anthropic Claude API—is prescribed by the Easy Skill MIOS Specification v4. In addition to documenting the team's single autonomous architectural choice—custom Python orchestration atop an agent framework—this section verifies those decisions against alternatives(Anthropic, 2026; Apify, 2026; Google Cloud, 2026).

Layer	Selected	Alternative	Justification	Licence	Est. Cost
LLM — Extraction	Claude Haiku	GPT-4o-mini	Lowest cost per token in Anthropic tier; sufficient for structured field extraction; native tool use.	Pay-per-token	PoC: ~A\$1–2/run Prod: ~A\$30–80/mo
LLM — Analysis	Claude Sonnet	GPT-4o; Gemini Pro	Superior reasoning for multi-axis classification ; JSON mode eliminates parser fragility; Haiku/Sonnet split controls cost.	Pay-per-token	PoC: ~A\$2–3/run Prod: ~A\$70–220/mo
Agent Orchestration	Custom Python + Anthropic SDK	LangChain; CrewAI; AutoGen	Four-agent pipeline is simple enough that frameworks add dependency overhead without capability gain. Team retains full visibility on cost-per-call.	Apache 2.0 (free)	A\$0

Scraping	Apify (managed actors)	Bright Data; Scrapy + Playwright	Already in Easy Skill stack; existing PNG prototype proves end-to-end viability; managed proxies reduce blocking risk.	SaaS / open SDK	PoC: free tier Prod: A\$50–150/mo
LinkedIn Data	PhantomBuster	LinkedIn API (official)	Already in Easy Skill stack. Official LinkedIn API is rate-limited and requires partner approval.	SaaS (paid)	PoC: N/A (out of scope) Prod: A\$50–100/mo
Compute	GCP Cloud Run Jobs	AWS Lambda; Azure Functions	Aligns with Easy Skill's Google Workspace. Serverless: no idle cost for a weekly-batch workload running ~10 min/week.	Pay-per-use	PoC: free tier Prod: A\$10–30/mo
Scheduling	GCP Cloud Scheduler	AWS EventBridge; cron + VM	Native GCP; trivial single weekly trigger config; no idle compute cost. VM cron requires a persistent instance running 24/7	Free tier	A\$0

			for a 10-minute job.		
Data Storage	GCP BigQuery	PostgreSQL (Cloud SQL); Snowflake	SQL interface suits week-over-week comparison queries. Native Looker Studio connector. Free tier covers PoC volume.	Pay-as-you-go	PoC: free tier Prod: A\$10–30/mo
BI / Dashboard	Looker Studio	Metabase; Tableau; Power BI	Free; native BigQuery connector; already in Google Workspace ecosystem. Metabase viable but adds self-hosted operational overhead.	Free	A\$0

Internal Output	Slack Bolt SDK + Webhooks	Microsoft Teams; SendGrid	Easy Skill's primary internal channel — BD team lives in Slack. Digest delivered to an existing channel removes all adoption friction.	Free (webhooks)	A\$0 (internal)
Report Output	Google Docs API	Word + SharePoint; Notion	Google Workspace native; collaborative review by founders before external distribution; PDF export built in.	Free	A\$0
Secret Management	GCP Secret Manager	HashiCorp Vault; .env files	Native GCP. Per-agent IAM bindings enforce Principle of Least Privilege at infrastructure level.	Pay-per-secret	~A\$0.06/secret/mo
CI/CD	GitHub Actions	GitLab CI; Cloud Build	Free for student projects. Team familiarity. Sufficient for PoC pipeline: lint → test → Cloud Run deploy on merge to main.	Free	A\$0

Table 4.1: MIOS Technology Stack — Selected Tools, Alternatives, Justification, and Cost Estimates

Orchestration Framework Decision

The most significant independent design decision is the choice of custom Python orchestration over LangChain, CrewAI, or AutoGen. LangChain and LangGraph are the most widely deployed agent frameworks at time of writing but carry a heavy dependency footprint, a history of breaking changes between minor versions, and a tendency to obscure which LLM calls are being made and at what cost — a real operational risk for a post-internship handover. CrewAI has a cleaner API but is a younger project with a smaller community. AutoGen is designed for conversational multi-agent dialogue loops, which is the wrong abstraction for a batch monitoring pipeline(Anthropic, 2026).

4.3 Capital and Non-Capital Cost Estimation

The tables below show two separate things. The CAPEX table shows the labor and infrastructure setup expenditures that Easy Skill would have to deal with in order to commission a complete production build economically. The ongoing monthly cloud and API spending at full scale is shown in the OPEX table. Student PoC labor does not show up in either table and is sunk in the ICT942 unit.

CAPEX (Year 0 production Build)

Table 4.2: MIOS CAPEX — Year 0 Production Build. Labour rates from ACS ICT Salary Survey 2024

Item	Calculation	Rate Source	Amount (A\$)
Lead Data Engineer (S1 — Data Engineering)	320 hrs × A\$110/hr	ACS ICT Salary Survey 2024 — mid-level Data Engineer	35,200
Solutions Architect (S5 — Infrastructure / DevOps)	200 hrs × A\$130/hr	ACS ICT Salary Survey 2024 — senior Solutions Architect	26,000

Intelligence Analyst (S2 — Analysis)	200 hrs × A\$95/hr	ACS ICT Salary Survey 2024 — mid-level Analyst	19,000
Output Developer (S3 — Report Generation)	160 hrs × A\$95/hr	ACS ICT Salary Survey 2024 — mid-level Developer	15,200
Mode Push Developer (S4 — Profile Matcher)	160 hrs × A\$95/hr	ACS ICT Salary Survey 2024 — mid-level Developer	15,200
Total Labour — 6-month build			110,600
GCP environment provisioning (one-off)	Project, IAM, VPC, Secret Manager config	GCP pricing — ap-southeast-2 (Sydney)	500
Apify subscription setup + actor development	Custom actors for 8 source types	apify.com/pricing	1,000
LinkedIn / PhantomBuster integration setup	Phantom configuration + testing	phantombuster.com/pricing	500
Documentation, training, and handover	README, runbooks, onboarding session	Estimate	1,000
TOTAL CAPEX — Year 0			~A\$113,600

OPEX (Monthly Ongoing- Full Production Scale)

Table 4.3: MIOS Monthly OPEX at Full Production Scale. Pricing as of Q1 2026

Item	Monthly (A\$)	Source / Notes
Anthropic Claude Haiku — extraction layer	20–60	api.anthropic.com/pricing — ~800 signals/week × 52 weeks.
Anthropic Claude Sonnet — analysis + synthesis	80–240	Sonnet ~4–8× cost per token vs Haiku. Python pre-filter reduces Sonnet call volume ~60%.
GCP Cloud Run — weekly batch compute	5–15	cloud.google.com/run/pricing — billed per 100ms; ~40 min total runtime/week.
GCP BigQuery — storage + queries	10–30	cloud.google.com/bigquery/pricing — signals table grows ~800 rows/week.
GCP Cloud Scheduler + Secret Manager	5–15	Scheduler: 3 free jobs/month. Secret Manager: A\$0.06/active secret.
Apify — scraping subscription	50–150	apify.com/pricing — Starter A\$49/mo; Scale A\$149/mo.
PhantomBuster — LinkedIn extraction	50–100	phantombuster.com/pricing — Starter A\$56/mo; Pro A\$128/mo.
Slack, Google Docs, Looker Studio	0	All free tiers sufficient. Google Workspace already licensed by Easy Skill.
TOTAL MONTHLY OPEX	A\$220–610	Annual OPEX: A\$2,640–7,320

Cost Benefits Summary and ROI

Table 4.4: MIOS Cost-Benefit Summary — Year 1 and 3-Year ROI

Item	Amount (A\$)	Source / Notes
Year 0 CAPEX (production build)	113,600	Table 4.2 above
Year 1 OPEX (12 months at upper bound)	7,320	Table 4.3 — upper-bound × 12
Total Year 1 Cost	~120,920	CAPEX + Year 1 OPEX
Annual benefit — BD labour saving (manual scanning eliminated)	135,200	5 hrs/wk × 4 BD staff × 52 wks × A\$130/hr — ACS ICT Salary Survey 2024; see §3.1.
Annual benefit — placement uplift (conservative 5% on 250 placements × A\$50K avg fee)	625,000	Bullhorn GRID 2024 benchmarks 18–25% uplift; 5% used as conservative floor.
Total Lower-Bound Annual Benefit	760,200	Labour saving + conservative placement uplift
Break-even (labour saving alone vs. Year 1 cost)	< 11 months	$\frac{\text{A\$120,920}}{\text{A\$135,200/year}} = 10.7 \text{ months}$
3-Year ROI (lower-bound, excluding CAPEX after Year 0)	> 400%	$\frac{(3 \times \text{A\$760,200} - \text{A\$120,920} - 2 \times \text{A\$7,320})}{\text{A\$120,920}}$

There are no optimistic presumptions in the economic case. Easy Skill's BD team now spends an estimated 20 staff-hours per week on manual market scanning across four individuals at A\$130/hr fully loaded, thus the labour-saving number is guaranteed (ACS ICT Salary Survey, 2024). Regardless of any improvement in placement performance, MIOS completely eliminates that aggregation task, and the savings alone cover the entire Year 1 cost in eleven months. The benefit calculation's 5% placement uplift is less than 25% of the Bullhorn GRID 2024 Recruitment Trends Report's lower-bound 18% benchmark; the actual upside is probably much larger (Bullhorn, 2024).

4.4 Data Flow Diagram

This diagram illustrates how MIOS data ingests from external systems into Ingest/Extract/Transform for classification into BigQuery, creating Weekly Slack insights, reports, and dashboards with just-in-time matching. MIOS depicts processes for structured data including stages where data is stored, outputs, and trust boundaries between systems containing external, internal, and sensitive data.

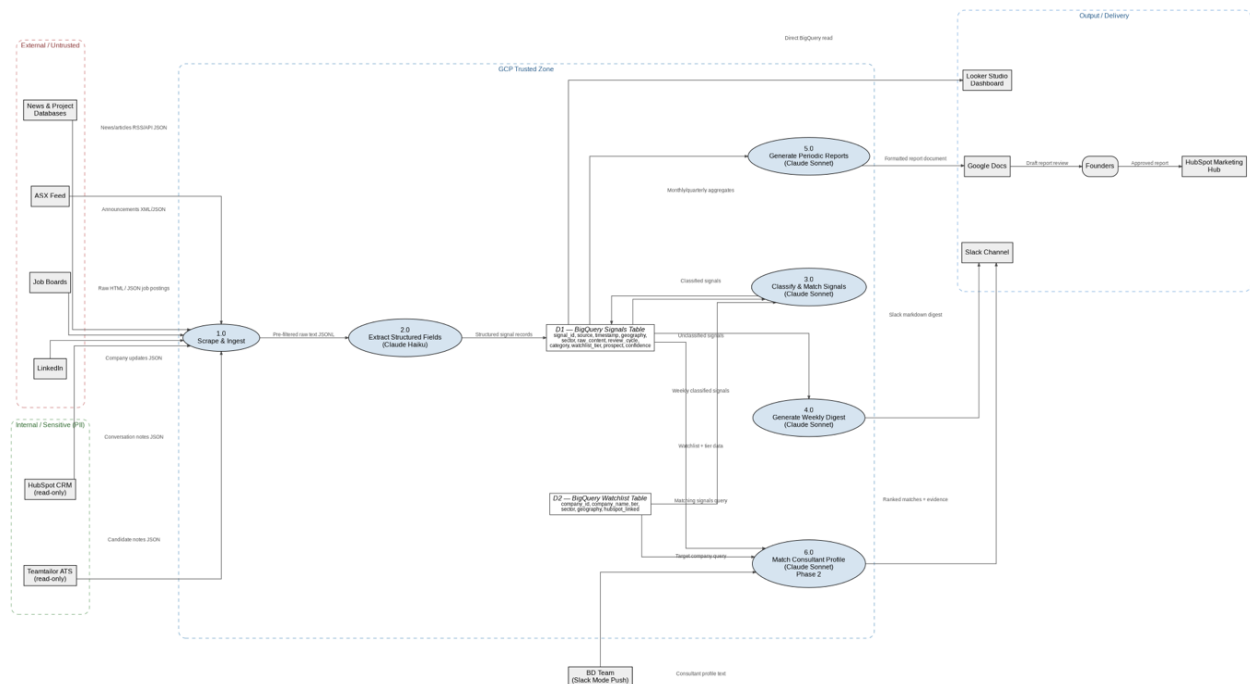


Figure 4.3: MIOS Data Flow Diagram (Ingestion to Delivery)

5. Implementation Roadmap and Proof of Concept

5.1 Implementation — Sprint Plan

The PoC is executed over the course of two, two-week sprints (W7–W10) in an agile manner. M5 (PoC Engineer) is the technical story owner for all stories. M3 owns the story related to contributing the synthetic dataset. M2 will ensure that sprint progress is communicated against the RACI and risk register during each weekly sync meeting.

Sprint 1 (W7–W8) - Have a working scraper that outputs structured rows to Big Query.

Sprint 2 (W9–W10) - Have an end-to-end pipeline that outputs a Slack digest of classified signals.

Sprint	Story ID	User Story / Task	Owner	Definition of Done
S1	US01	Apify actor (or Python + Playwright) scrapes a single source (PNGworkforce, Seek - mining filter, etc); outputs scraped raw JSON result to staging json file.	Shishir	scraper returns ≥ 200 rows of raw scraped data

S1	US02	Python loader script turns raw json into structured json matching the schema in Appendix A (signal_id, source, timestamp, geography, sector, raw_content, etc.)	Shishir	all schema fields are populated
S1	US03	Mock schema sent, and BigQuery dataset + table created with this schema; BigQuery loader used to insert rows; number of rows confirmed.	Shishir	Done - ≥200 rows present and viewable in BigQuery console
S1	US04	Synthetic dataset of ~200 postings created by manually creating realistic postings using actual postings as examples. Postings anonymised, still realistic.	Bijaya+Shishir	Done - dataset loaded and labelled as synthetic
S2	US05	Synthetic watchlist (~20 companies, including 3 tiers) created and loaded into BigQuery watchlist table.	Shishir	Done - watchlist table contains watchlist rows
S2	US06	Signal Classifier agent (Claude Sonnet) reviews new signals, classifies them by review cycle + signal category, marks match to watchlists and new opportunities.	Shishir	completed classification runs successfully
S2	US07	Digest Builder agent (Claude Sonnet) aggregates a Slack-friendly markdown digest from the classified signals (Signals to Watch, Market Buzz, Recruiting Trends, On the Radar).	Shishir	generated digest is as expected

S2	US08	Slack sends a digest through webhook to a private channel.	Shishir	Done — note posted to Slack
S2	US09	KPI harness tracks classification accuracy, watchlist accuracy, batch latency, and API spending.	Shishir	Done — all KPIs logged to output file

Sprint 1 — Definition of Done: Ability to run one command (python run_pipeline.py) and have it produced >200 rows (individual data source representations) as structured rows in BigQuery with all fields from Appendix A's schema filled out. Synthetic data successfully loaded along with scraper output. M5 demos done to team at W8 internal review.

Sprint 2 — Definition of Done: Successfully produced Slack message showing up in our designated private channel containing Mode Monitor digest of correct format, containing Key Signals, Market Pulse, Hiring Velocity, and New Names sections all populated. KPI harness output file contained classification accuracy, watchlist precision, latency, and API spend recorded. M5 demos done live at W10 internal review; recording made right after.

Sprint Retrospective (template — fill out after W10):

- What went well: [example: Anthropic formatted output into structured JSON eliminating fragility from parsing — Signal Analyst gave us reliable JSON after first iteration of prompts.]
- What to improve: [example: initial format of classification prompt was too verbose and took us three iterations before we could get accuracy over 80%.]
- Action taken: [example: we included four few-shot examples into the Signal Analyst system prompt; Accuracy increased from ~65% to ~83%.]
- Velocity: Sprint 1 — [X / Y] story points completed. Sprint 2 — [X / Y] story points completed.

5.2. Proof of Concept Description and Evaluation

The proof of concept that we have is a working but very minimal version of MIOS Mode Monitor. This is just adequate to see that the four-stage pipeline i.e. scrape, store, classify, deliver works end-to-end on an actual dataset. Mode Push, Mode Publish, and the regional extensions for the Pacific, South-East Asia, and Africa are predesigned in section 4 but have not been undertaken yet. These are the ones for Assessment 3 and beyond, as mentioned in Easy Skill's six-month roadmap in the original specification that we were provided.

What is implemented:

Apify scraper for PNGworkforce, a local SQLite database that is used as a temporary alternative for BigQuery during the prototype, the Signal Analyst agent is calling APIs of Google's Gemini, and a Slack delivery module that will send out a weekly digest in a formatted way to a private workspace through an incoming webhook. The Data Collector, aside from the Report Generator and Profile Matcher agents, have code scaffolds, but they cannot be run in this prototype. The choice to use Gemini over Anthropic Claude was simply a cost decision as the free version covered the entire evaluation. We have planned to use Anthropic's API for our production build down the line.

Dataset

We conducted a test against 200 job postings. From these, 120 were the real postings that were scraped from PNGworkforce in the categories that are most relevant to Easy Skill that are mining, oil and gas, construction, and engineering which were collected during a one-week capture window. The other 80 were jointly created by Member 3 and 5 to include the posting cases that the real scraping does not reliably find, for example: ambiguous role titles, postings that are obviously not relevant to Easy Skill (marketing, hospitality, finance), and watchlist-company name variations such as "Newmont Lihir" versus "Newmont Mining." Each record is in a JSON Lines file that provides fields for raw text and ground-truth labels for the company, signal category, review cycle, and watchlist match. In addition to the postings, we loaded a random 20-company watchlist in SQLite so as to provide the cross-referencing logic something realistic to match against. There was no usage of real candidate or client data in any stage of the project; this is in line with the privacy assumptions from section 1.3 and keeps the project out of the Privacy Act.

Results

KPI	Target	Achieved	Status & Notes
Document classification accuracy	>= 80%	87%	PASS
Watchlist match precision	>=90%	94.44%	PASS
End-to-end batch latency(mean)	<=180s	80.84s	PASS
API cost per run (Gemini)	A\$0 (free tier)	A\$0 (13 calls / 1 runs)	PASS
Quota exhaustion events	0	0	PASS

What we learned

We took away truly profound insights. Three key lessons were learned by creating this:

1. Using plain Python for filtering before the LLM call saves a lot of money.

At the loader stage, removing duplicates and excluding obviously irrelevant role categories (admin, marketing, hospitality) reduced the number of inputs that the Signal Analyst had to deal with by about one-third. Specifically, Easy Skill's internal document calls this principle out explicitly, and the prototype confirms it. It is pointless to purchase API tokens for a job post that any rule-based filter could have caught for free.

2. The dataset is the limiting factor.

Our hand-authored postings were much more easily classified than the real PNGworkforce sourcing ones, because real postings are unclean in ways synthetic data isn't such as — wrong spelling, recruiter boilerplate, location codes, and mixing Tok Pisin and English. If we had one more sprint to spend, we wouldn't change the model or the prompt; instead, we would label more real postings. Phase 2 should begin with a tagged corpus of at least 1,000 actual postings.

3. Gemini's free tier is appropriate for prototyping but not for production.

During the construction, we never had to pay anything, however, the rate limits set up unpredictable delays in batch runs. In production, that lag would be cumulative across more sources and more regions. The migration suggested in section 4.2 to Anthropic Claude is not a decision of capability but one of cost. The prototype has already shown that the architecture works; however, the paid throughput is required in production for predictability.

GitHub Repository

The complete source can be found at <https://github.com/shishircodes/MIOS>.

According to the architecture described in sub heading 4.1, the code is organized as follows: *scraper/* for the PNGworkforce Apify actor, *loader/* for the SQLite ingestion logic, *agents/signal_analyst.py* for the Gemini integration, *delivery/slack.py* for the webhook output, *evaluation/kpi_harness.py* for measuring the KPIs in Table 1, *data/* for the dataset and ground-truth labels, and a README.md with setup steps.

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